CDA 106
Numerical Methods for Data Science

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1) Gauss-Hermite 2 point method.

$$\int_{-\infty}^{\infty} e^{-x^2} f(x)\,dx = \frac{\sqrt{\pi}}{2}\left[f\left(-\frac{1}{\sqrt{2}}\right) + f\left(\frac{1}{\sqrt{2}}\right)\right]$$

2) Gauss Hermite 3 point method.

$$\int_{-\infty}^{\infty} e^{-x^2} f(x)\,dx = \frac{\sqrt{\pi}}{6}\left[f\left(-\frac{\sqrt{6}}{2}\right) + 4f(0) + f\left(\frac{\sqrt{6}}{2}\right)\right]$$

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi} \cong 1.7724$$

we solved.

Example $I = \int_{-\infty}^{\infty} e^{-x^2} \underbrace{\frac{1}{1+x^2}}_{f(x)}\,dx$

use Gauss Hermite 2 point

$$f(x) = \frac{1}{1+x^2}$$

$$f\left(-\frac{1}{\sqrt{2}}\right) = \frac{2}{3} = f\left(\frac{1}{\sqrt{2}}\right)$$

$$= 0.6667$$

using formula,

$$I = \int_{-\infty}^{\infty} e^{-x^2} f(x)\,dx = \frac{\sqrt{\pi}}{2}\left[f\left(-\frac{1}{\sqrt{2}}\right) + f\left(\frac{1}{\sqrt{2}}\right)\right]$$

$$I = \frac{\sqrt{\pi}}{2}\left[\frac{2}{3} + \frac{2}{3}\right] = \frac{2\sqrt{\pi}}{3}$$

$$= 1.1816$$

Alternatively,

$$I = \frac{\sqrt{\pi}}{2}\left[0.6667 + 0.6667\right] = 0.8862$$

$$= 1.1816$$

Apply 3 point (Exercise)

Ans : 1.4179

Exercise Evaluate $I = \int_{-\infty}^{\infty} \frac{e^{-x^2}}{1+x+x^2}\, dx$

using i) Gauss Hermite 2 point rule

ii) Gauss Hermite 3 point rule

Ans i) 1.5192

ii) 1.4926

Summary

Numerical Integration

$$\int_a^b f(x)\,dx$$

Newton-Cotes

- x_i's were found
- x_i's were equidistant
- x_i's were fixed.
- More accurate result was obtained by increasing no of x_i's

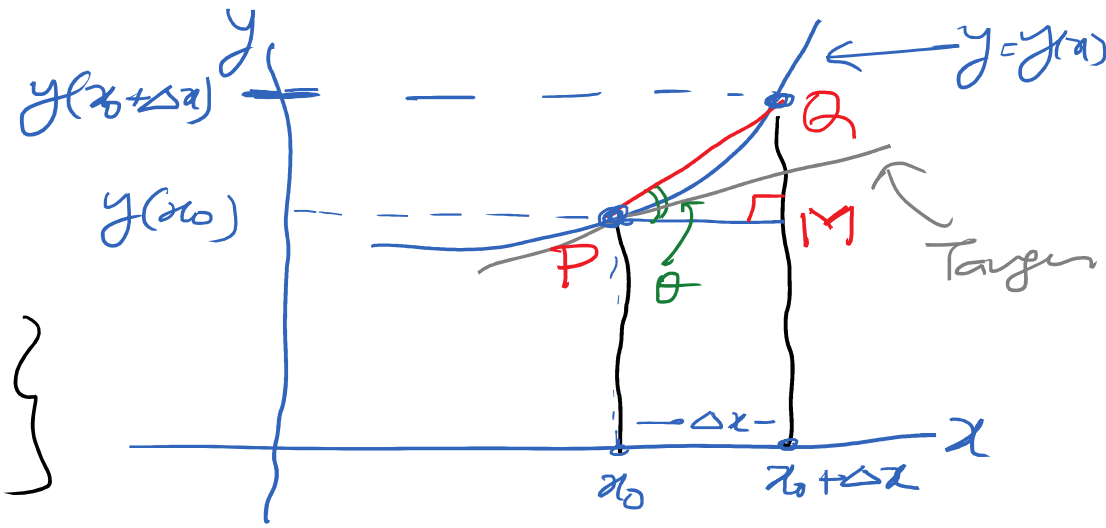
- Rectangular
- Trapezoidal
- Simpson's $\frac{1}{3}$
- Simpson's $\frac{3}{8}$

Gaussian

- x_i's were predefined
- x_i's were roots of some specific polynomials
- less functional evaluation needed

Numerical Differentiation

$y = y(x)$



Slope of the Chord PQ

$$= \frac{\Delta y}{\Delta x} \quad \left\{ \begin{array}{l} \text{Change of } y \\ \text{w.r.t. change of } x \end{array} \right\}$$

$\Delta x \to 0 \qquad Q \to P$

$$\boxed{\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}}$$

slope of the tangent

derivative

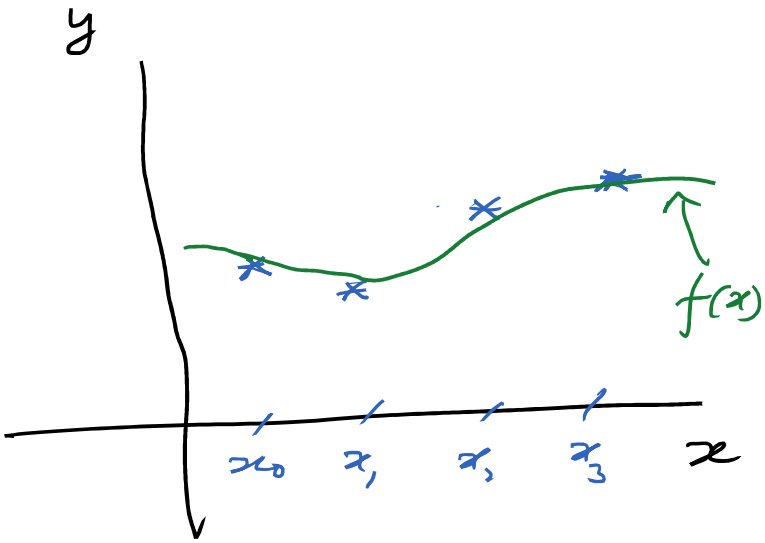
$PM = \Delta x$, $QM = y(x_0 + \Delta x) - y(x_0)$
$= \Delta y$

$$\tan\theta = \frac{QM}{PM} = \frac{\Delta y}{\Delta x}$$

slope of PQ (PM || to x-axis)

Consider we are given

x	x_0	x_1	x_2	x_3
$y=f(x)$	$f(x_0)$	$f(x_1)$	$f(x_2)$	$f(x_3)$



Suppose, you want to find derivative at any point

$\longrightarrow$ then, you will use Numerical differentiation.

How ??

One way to deal with this problem is by finding a "suitable function". Then take the derivative.

Forward difference Operator

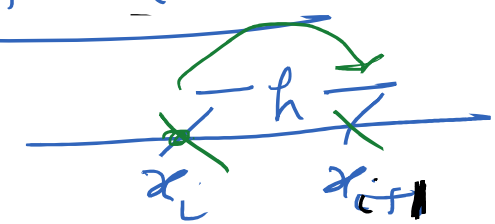
x_0	x_1	x_2	$\cdots$	x_n
$f(x_0)$	$f(x_1)$	$f(x_2)$	$\cdots$	$f(x_n)$

Data points are equidistant

$$x_1 - x_0 = x_2 - x_1 = x_3 - x_2 \cdots = x_n - x_{n-1} = h$$

$$x_{i+1} - x_i = h \quad \forall i$$

$$\Rightarrow \quad x_{i+1} = x_i + h$$



Define

$$\boxed{\Delta f(x_i) = f(x_{i+1}) - f(x_i)} \quad \forall i$$

or

$$\boxed{\Delta f(x_i) = f(x_i + h) - f(x_i)} \quad \forall i$$

Therefore,

$$\Delta^2 f(x_i) = \Delta\left(\Delta f(x_i)\right) = \Delta\left(f(x_i+h) - f(x_i)\right)$$

$$= \Delta f(x_i+h) - \Delta f(x_i) = f(x_i+2h) - f(x_i+h) - f(x_i+h) + f(x_i)$$

$$\Delta^2 f(x_i) = f(x_i + 2h) - 2f(x_i + h) + f(x_i)$$

In the similar fashion, we can calculate

$\Delta^3 f(x_i)$, $\Delta^4 f(x_i)$, ...

→ Why do we need them

A: For getting numerical diff via a ~~fact~~ function

Look at

$\boxed{h=1}$

x	$1 \sim x_0$	$2 \sim x_1$	$3 \sim x_2$
$f(x)$	2	6	12
	$f(x_0)$	$f(x_1)$	$f(x_2)$

$i = 0, 1, 2$

$$\Delta f(x_i) = f(x_{i+1}) - f(x_i) \quad \forall i$$

$$\Delta f(x_0) = f(x_1) - f(x_0) = 6 - 2 = 4$$

$$\Delta f(x_1) = f(x_2) - f(x_1) = 12 - 6 = 6$$

$$\Delta^2 f(x_0) = f(x_2) - 2f(x_1) + f(x_0) = 12 - 2\times 6 + 2 = 2$$

$$\boxed{\begin{aligned} &= \Delta\left(f(x_1) - f(x_0)\right) \\ &= \Delta f(x_1) - \Delta f(x_0) = 6 - 4 = 2 \end{aligned}}$$

Forward difference Table

Ex-1D

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$
$x_0 = 1$	$2 \to f(x_0)$	$4 \to \Delta f(x_0)$	
$x_1 = 2$	$6 \to f(x_1)$	$6 \to \Delta f(x_1)$	$2 \to \Delta^2 f(x_0)$
$x_2 = 3$	$12 \to f(x_2)$		

$$\boxed{h = 1}$$

Therefore, the polynomial function in example 1 D can be found.

$$P_2(x) = f(x_0) + \frac{x - x_0}{h} \Delta f(x_0) + \frac{(x-x_0)(x-x_1)}{2!\, h^2} \Delta^2 f(x_0)$$

$$= 2 + \frac{x-1}{1} \times 4 + \frac{(x-1)(x-2)}{2 \times 1^2} \times 2$$

$$= 2 + 4x - 4 + x^2 - 3x + 2$$

$$\boxed{P_2(x) = x^2 + x}$$

Example Construct Forward difference Table for

$\boxed{h = 1}$

x	1	2	3	4	5
$f(x)$	1	8	27	64	125

	x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
x_0	1	1 $f(x_0)$				
			7 $\Delta f(x_0)$			
x_1	2	8 $f(x_1)$		12 $\Delta^2 f(x_0)$		
			19		6 $\Delta^3 f(x_0)$	
x_2	3	27 $f(x_2)$		18		0 $\Delta^4 f(x_0)$
			37		6	
x_3	4	64 $f(x_3)$		24		
			61			
x_4	5	125 $f(x_4)$				

Let us construct polynomial for above data.

$$P_3(x) = f(x_0) + \frac{x-x_0}{h}\Delta f(x_0) + \frac{(x-x_0)(x-x_1)}{2!\,h^2}\Delta^2 f(x_0)$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_2)}{3!\,h^3}\Delta^3 f(x_0) + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{4!\,h^4}\Delta^4 f(x_0)$$

$$= 1 + \frac{x-1}{1}\times 7 + \frac{(x-1)(x-2)}{2\times 1^2}\times 12 + \frac{(x-1)(x-2)(x-3)}{\cancel{6}\times 1^3}\times \cancel{6} + 0$$

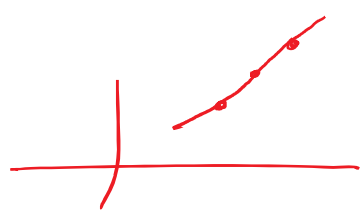
$$= 1 + 7x - 7 + 6x^2 - 18x + 12 + x^3 - 6x^2 + 11x - 6$$

$$\boxed{P_3(x) = x^3}$$

To find the derivative in above case

i.e. $f'(1) = ?$

$$f'(1) = P_3'(1) = 3x^2\Big|_{x=1} = 3 \quad \text{Ans.}$$

— x —

$n+1$ data $\longrightarrow$ ~~fit~~ fits nth degree polynomial } Highest possible